

Windows System Administration and Troubleshooting Lab

Author:

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Role Focus:

IT Support • Help Desk • Desktop Support • Windows System Administration

Lab Objectives

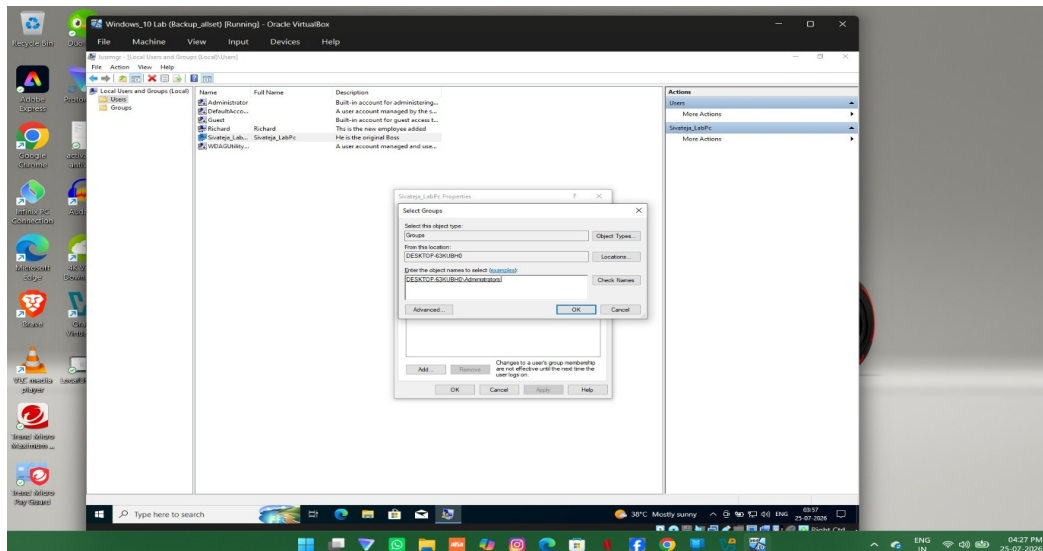
- Perform common Windows administration tasks.
- Manage local users and groups.
- Configure Windows system settings.
- Troubleshoot operating system issues.
- Use built-in administrative and diagnostic tools.
- Gain hands-on experience with Windows environments commonly used in enterprise IT.

Component	Details
Host Operating System	Windows 11 Pro
Guest Operating System	Windows 10 Pro
Virtualisation Platform	Oracle VirtualBox
Environment	Home Lab
Tools Used	Computer Management, Local Users & Groups, Control Panel, Registry Editor, Command Prompt
Platform	Desktop

Exercises:

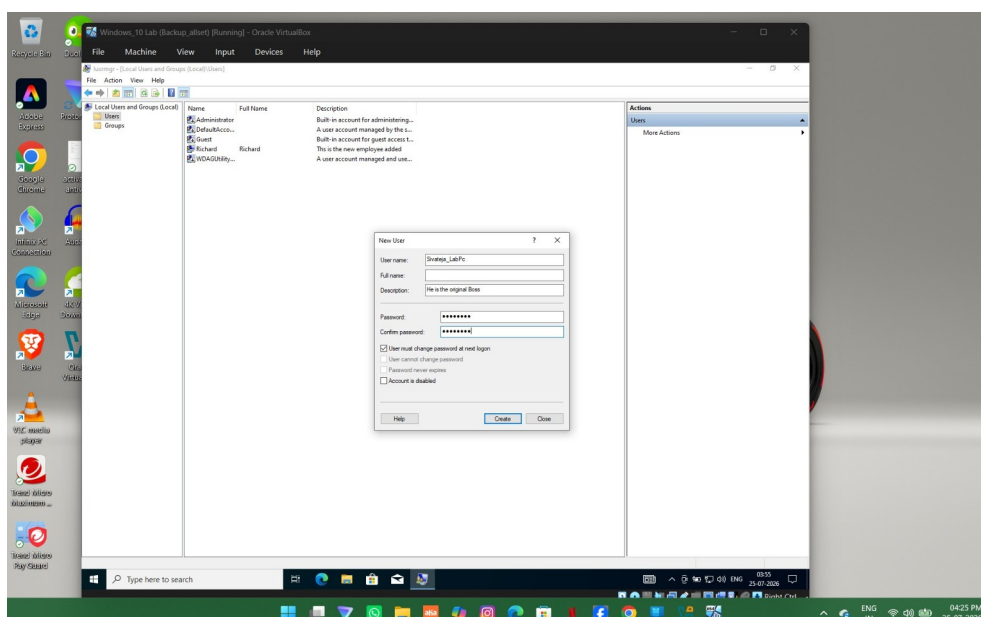
Assigning Administrative Privileges to a Local User

Created a standard local user account and added the user to the local **Administrators** group using **Local Users and Groups (lusrmgr.msc)**. This grants the user elevated privileges, allowing them to install software, modify system settings, manage other local accounts, and perform administrative tasks. Assigning administrator rights should be performed carefully and only when required, following the principle of least privilege.



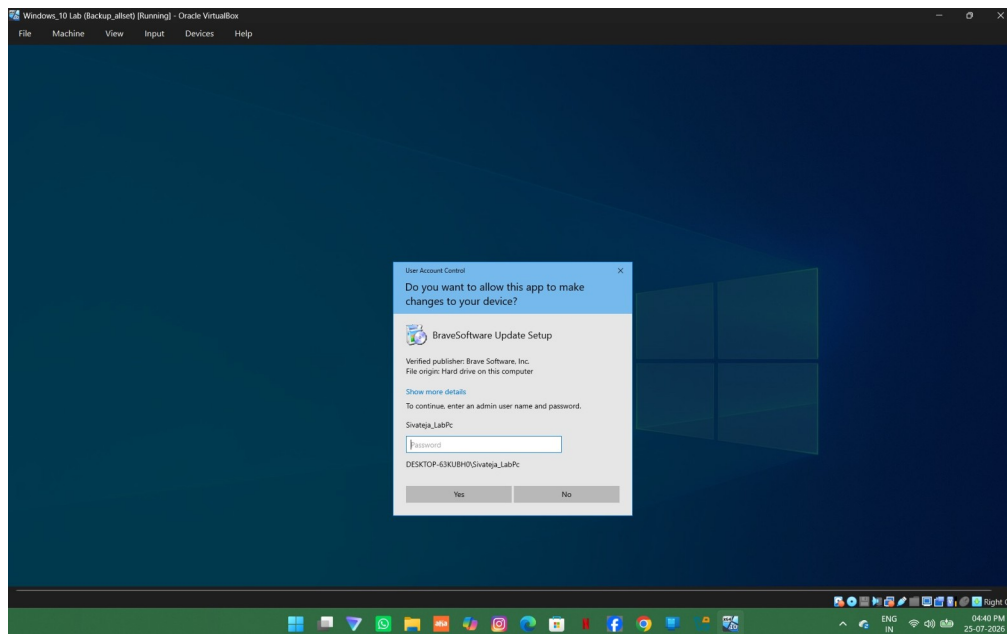
Creating a Local User Account

Created a new local user account using the Local Users and Groups (lusrmgr.msc) console. During account creation, a username, password, and description were configured, along with password policy options such as requiring the user to change the password at the next login. This demonstrates the process of provisioning local user accounts on standalone Windows systems where Active Directory is not available.



Verifying Administrative Privileges with User Account Control (UAC)

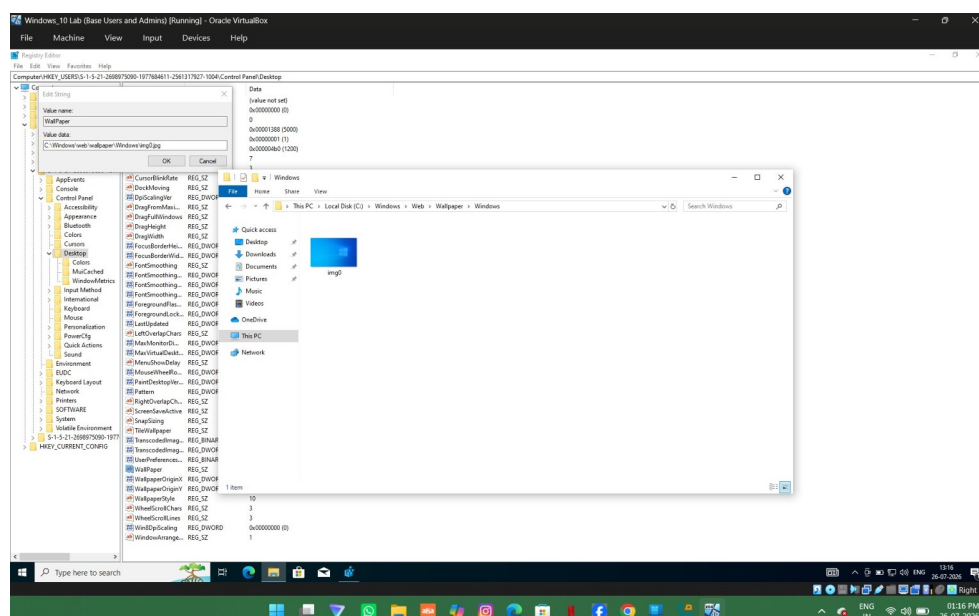
Verified that the assigned administrative privileges were successfully applied by attempting to perform an administrative action. Windows User Account Control (UAC) prompted for administrator credentials before allowing the operation to continue, confirming that the account possesses elevated privileges while still protecting the system from unauthorised changes. UAC provides an additional security layer by requiring explicit approval before administrative tasks are executed.



Modifying Desktop Wallpaper Configuration Using Registry Editor

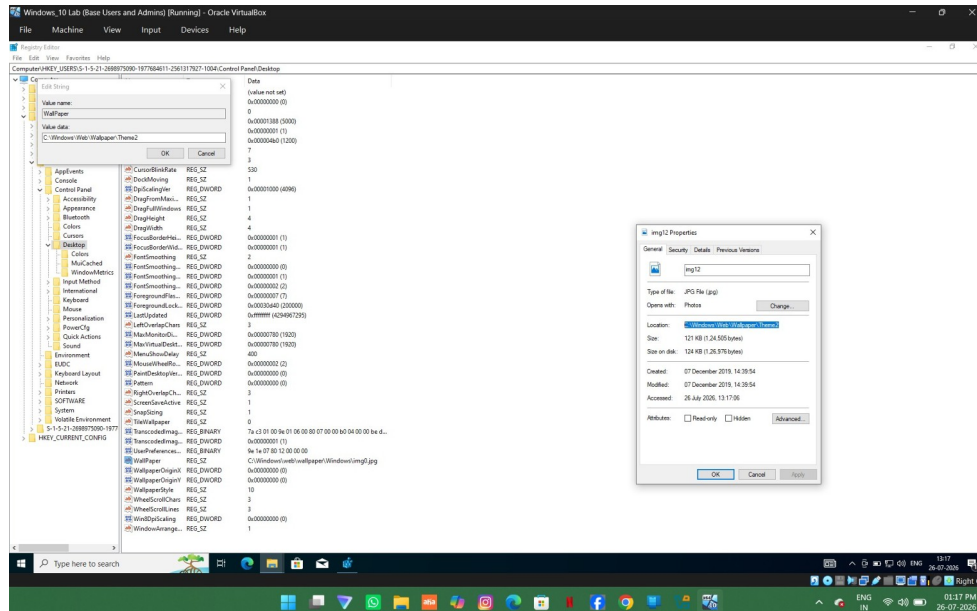
Navigated to the user's Registry hive to locate the desktop wallpaper configuration under: HKEY_CURRENT_USER\Control Panel\Desktop

Verified the Wallpaper registry value, which stores the absolute path of the current desktop background image. The configured wallpaper location was cross-checked with the corresponding file in the Windows wallpaper directory to confirm that the registry entry matched the actual image stored on disk.



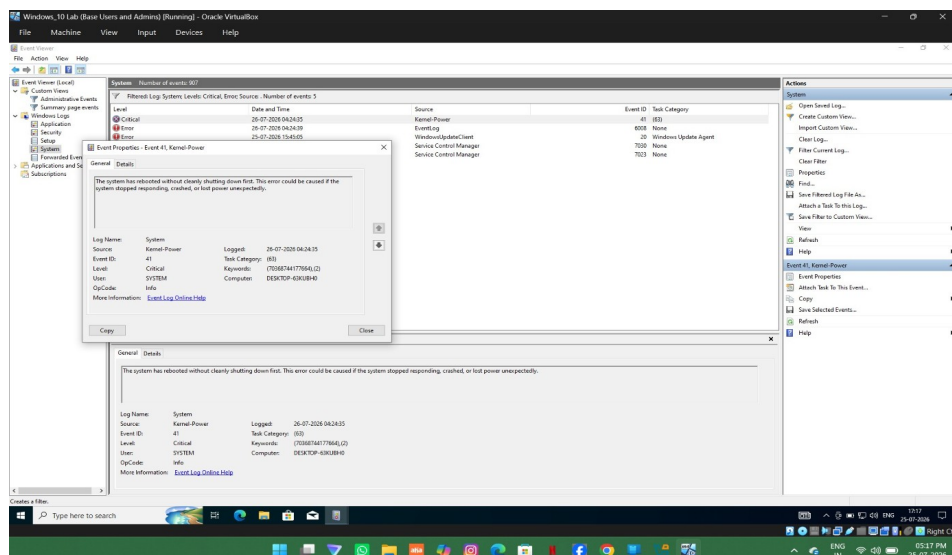
Updating the Desktop Wallpaper Registry Path

Modified the Wallpaper registry value to reference a different image stored on the local system. This demonstrates how desktop personalisation settings are maintained within the Windows Registry. Since Windows caches wallpaper settings, the change does not immediately update the desktop background and typically requires the user to sign out, restart Windows Explorer (explorer.exe), or reboot the system for the new wallpaper to be applied.



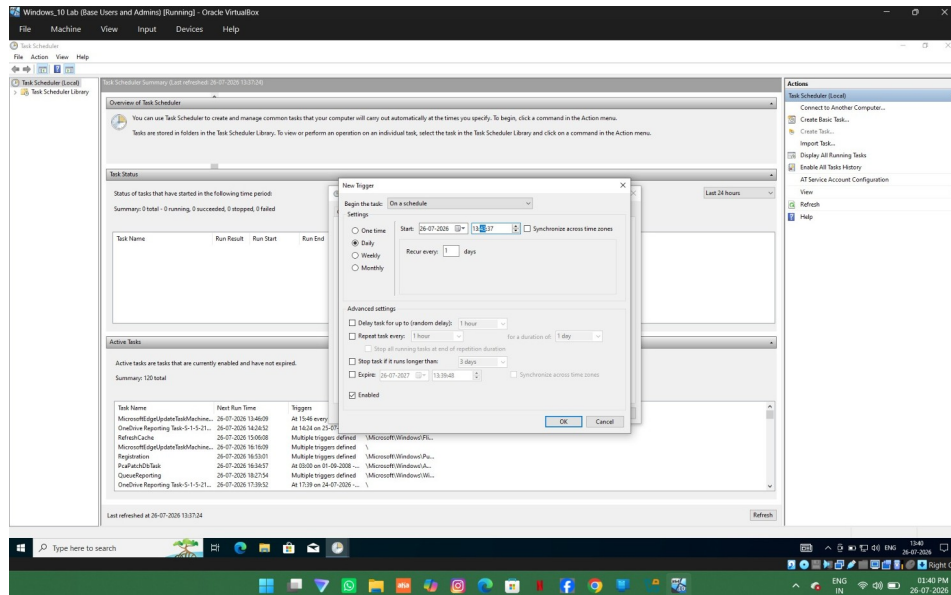
Using Event Viewer for System Troubleshooting

Used Event Viewer (eventvwr.msc) to review Windows System Logs and identify critical system events. Applied log filtering to display only Critical and Error events, allowing faster identification of issues affecting system stability. During the investigation, a Kernel-Power (Event ID 41) event was examined, indicating that the system restarted without a clean shutdown. Event Viewer provides detailed diagnostic information that assists IT Support professionals in troubleshooting unexpected shutdowns, application failures, driver issues, hardware faults, and operating system errors.



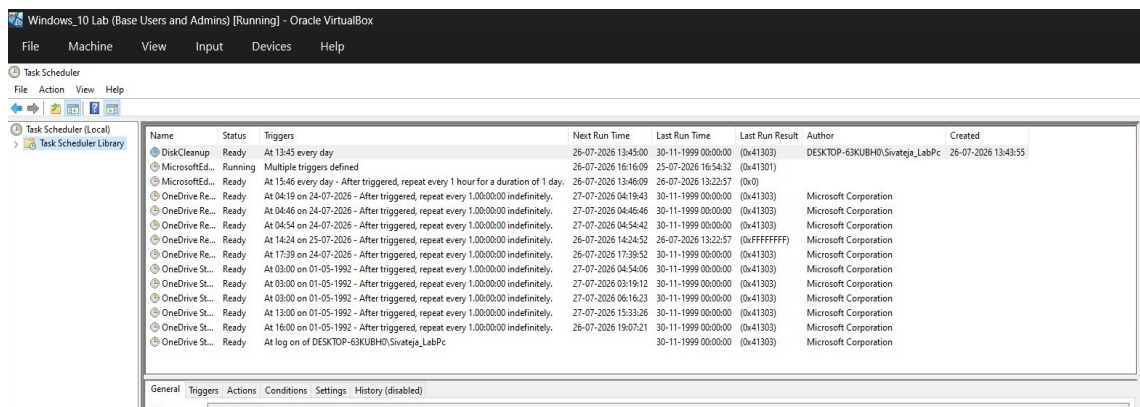
Creating a Scheduled Maintenance Trigger

Configured a scheduled trigger using Windows Task Scheduler to automate a maintenance task. The trigger was configured to run on a daily schedule at a specified time, demonstrating how administrators can automate recurring system operations without requiring manual intervention. Task Scheduler supports various trigger types including one-time, daily, weekly, monthly, system startup, user logon, and custom event-based execution.



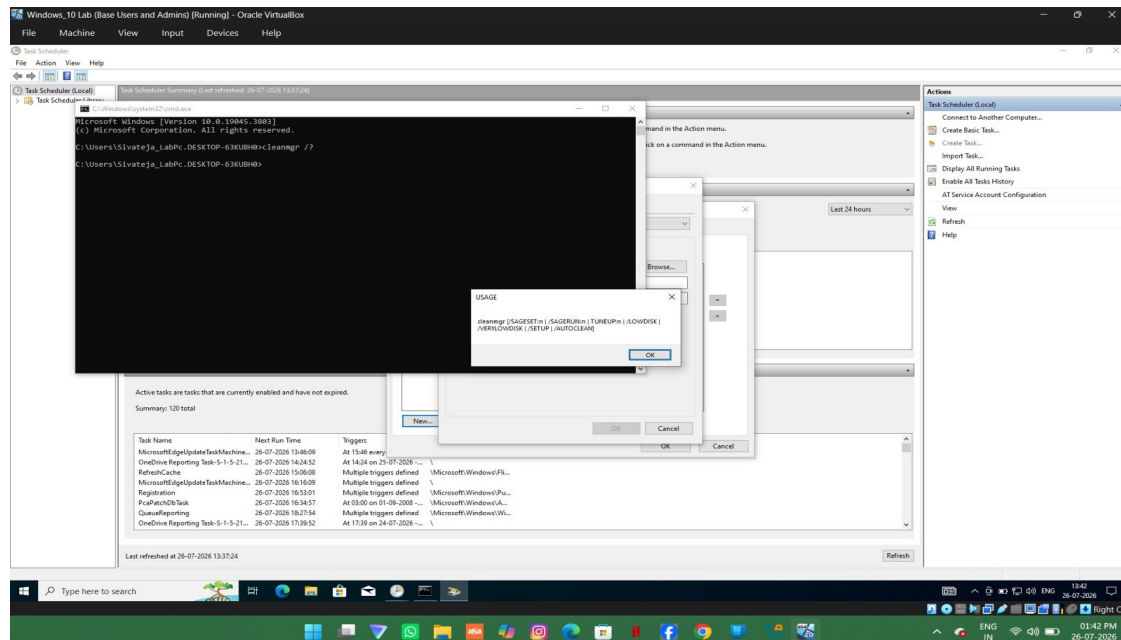
Automating System Maintenance with Task Scheduler

Used Windows Task Scheduler to create and manage a scheduled maintenance task. The task was configured to automatically execute Disk Cleanup (cleanmgr.exe) on a recurring schedule, demonstrating how routine administrative tasks can be automated to reduce manual intervention. Task Scheduler is widely used by system administrators to automate maintenance, monitoring, backups, updates, and custom scripts across Windows systems.



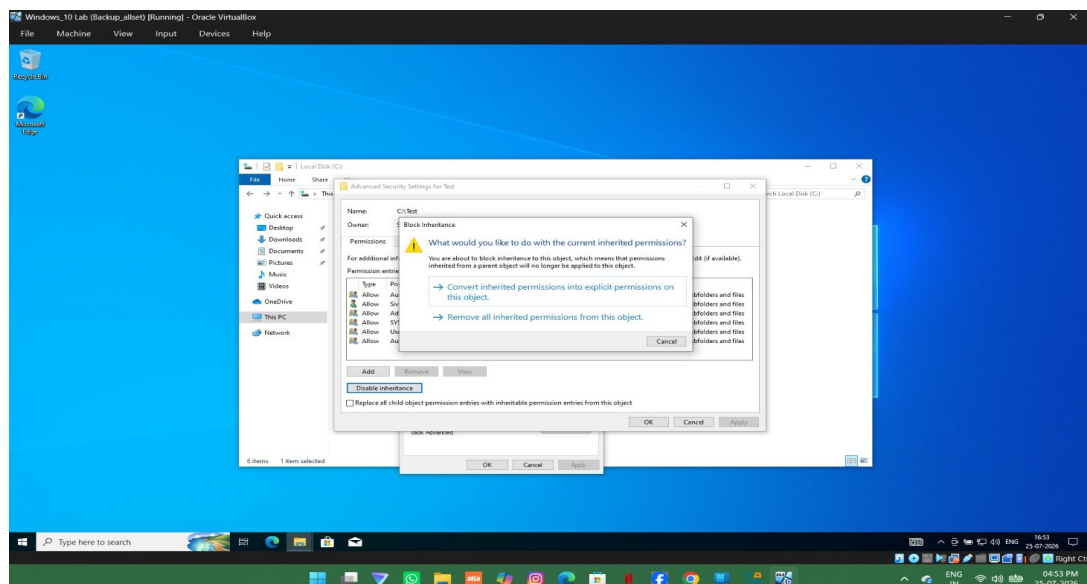
Identifying the Executable Used by a Scheduled Task

Verified the executable associated with the scheduled task by examining the command executed by Disk Cleanup (cleanmgr.exe). Reviewing the executable and its available command-line parameters helps administrators understand how scheduled tasks function, troubleshoot automation issues, and customize maintenance operations. This process demonstrates how Task Scheduler integrates with Windows command-line utilities to automate system administration tasks.



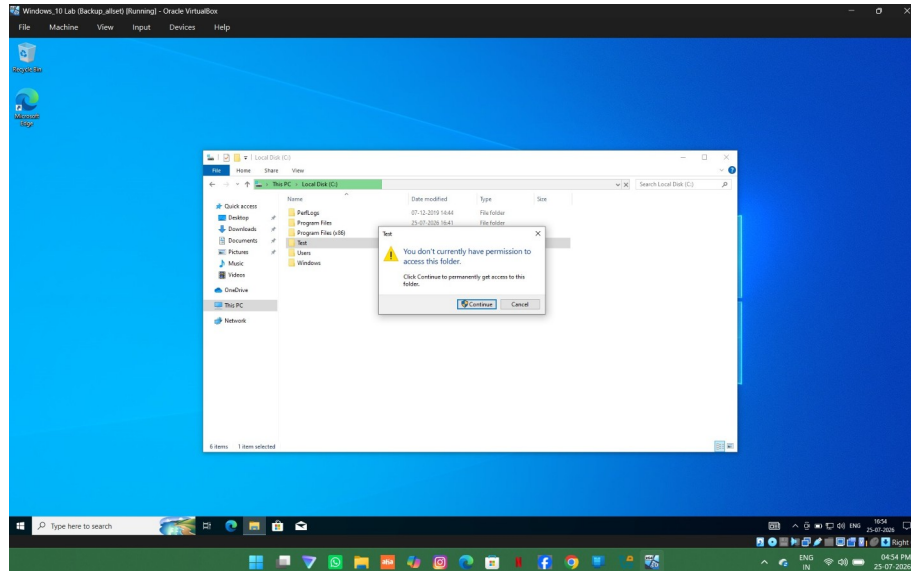
Configuring Folder Permission Inheritance

Modified the folder's Advanced Security Settings by disabling permission inheritance. Instead of automatically inheriting permissions from the parent directory, the inherited Access Control Entries (ACEs) were converted into explicit permissions. This allows administrators to create a custom permission model for sensitive folders without affecting parent directory permissions, providing greater control over access management.



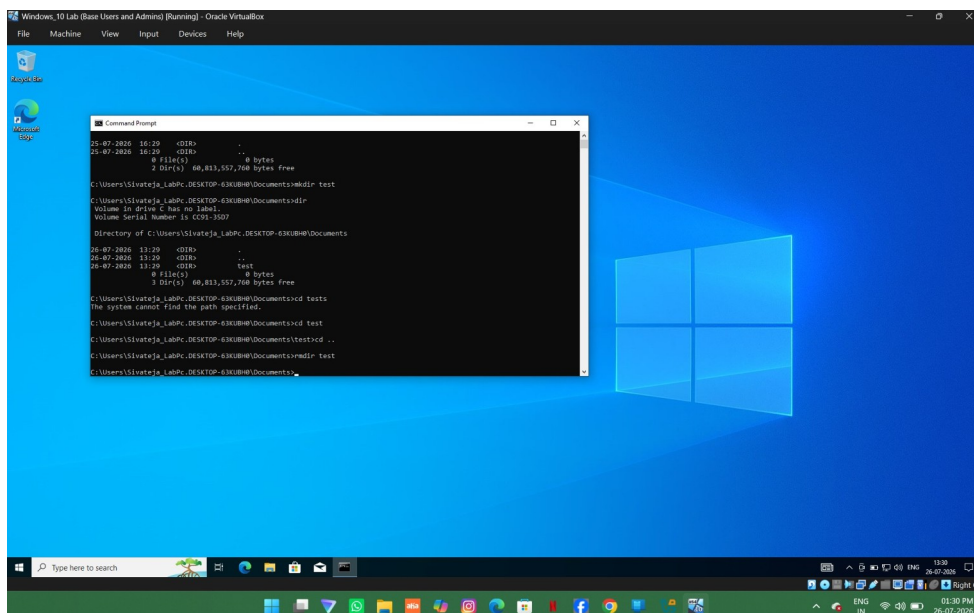
Restricting Folder Access Using NTFS Permissions

Configured NTFS permissions so that only authorised administrative users could access the protected folder. When a standard user attempted to open the directory, Windows displayed an Access Denied prompt requesting administrative approval, demonstrating how NTFS permissions enforce least-privilege access and protect sensitive resources from unauthorised users.



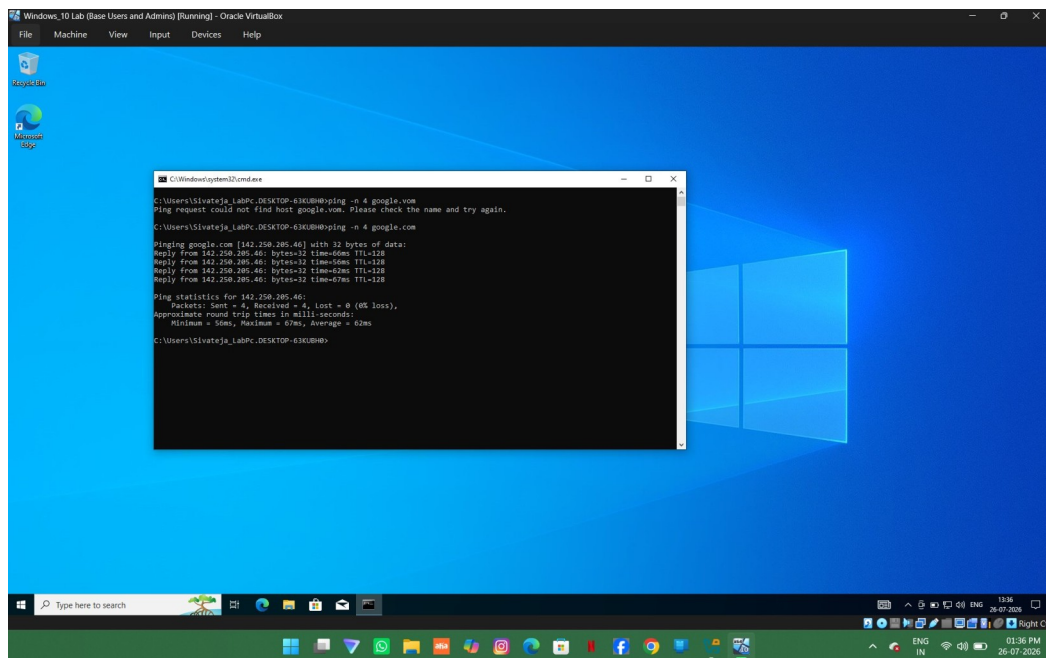
Basic File and Directory Management Using Command Prompt

Used Windows Command Prompt (cmd.exe) to perform basic file system navigation and directory management. Commands such as mkdir, dir, cd, and rmdir were used to create, list, navigate, and remove directories. An invalid directory change was also attempted, demonstrating how Windows returns an error when the specified path does not exist. These commands form the foundation of Windows command-line administration and are commonly used for troubleshooting, scripting, and system management.



Testing Network Connectivity Using Ping

Used the ping command in Windows Command Prompt to verify network connectivity and DNS name resolution. An initial attempt to ping an incorrectly typed hostname demonstrated a DNS resolution failure, followed by a successful test using the correct domain (google.com) with the -n 4 parameter to send four ICMP Echo Requests. The results confirmed successful connectivity by displaying packet statistics, response times, and zero packet loss, illustrating one of the most fundamental network troubleshooting techniques used by IT support professionals.



The screenshot shows a Windows 10 desktop with a blue background. A Command Prompt window is open, displaying the following text:

```
C:\Windows\system32\cmd.exe
C:\Users\Sivateja_LabPC\DESKTOP-63KUBH>ping -n 4 google.com
Ping request could not find host google.com. Please check the name and try again.
C:\Users\Sivateja_LabPC\DESKTOP-63KUBH>ping -n 4 google.com
Pinging google.com [142.250.285.46] with 32 bytes of data:
Reply from 142.250.285.46: bytes=32 time=50ms TTL=128
Reply from 142.250.285.46: bytes=32 time=50ms TTL=128
Reply from 142.250.285.46: bytes=32 time=62ms TTL=128
Reply from 142.250.285.46: bytes=32 time=67ms TTL=128

Ping statistics for 142.250.285.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 50ms, Maximum = 67ms, Average = 62ms
C:\Users\Sivateja_LabPC\DESKTOP-63KUBH>
```

Conclusion:

The exercises covered local user and group management, privilege assignment, User Account Control (UAC), Registry Editor configuration, Event Viewer analysis, Task Scheduler automation, NTFS file permissions, Command Prompt administration, and basic network diagnostics.

Through this hands-on environment, administrative tasks were completed using both graphical management consoles and command-line utilities, reinforcing core Windows operating system concepts and troubleshooting methodologies. These practical exercises provide a solid foundation for supporting Windows endpoints in enterprise environments. Windows administration encompasses a broad range of technologies beyond the scope of this lab.